

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Models the motion of the ball by forming an equation of motion	AO3.1b	M1	$m \frac{d^2x}{dt^2} = (-12.5mx) \times 2$
	Uses SHM equations to form model for displacement	AO3.1a	M1	$\Rightarrow \frac{d^2x}{dt^2} = -25x$ $\therefore x = A\sin(5t)$
	Uses initial condition to find the constant	AO1.1a	M1	$\Rightarrow \dot{x} = 5A\cos(5t)$
	Obtains correct value for constant	AO1.1b	A1	
	Interprets 'their' value to find minimum distance from P	AO3.2a	A1F	when $t = 0$, $\dot{x} = 0.75$ so $0.75 = 5A$ $A = 0.15$ Hence $x = 0.15\sin(5t)$ Max displacement = 0.15 metres from O, when $\sin(5t) = \pm 1$, so minimum distance from P is 0.75 metres
(b)	Identifies a correct limitation of the model for example friction between ball and the surface or damping effect due to air	AO3.5b	B1	It is unlikely that the surface is perfectly smooth so friction will be acting. The ball will be likely to travel a smaller distance before coming to rest and the minimum distance of the ball from P may actually be greater than that calculated in part (a).
	Correctly infers whether the distance is too big or too small based on the limitation they have identified. Accept any well-reasoned inference.	AO2.2b	R1	
				Total 7 marks

Q2.

(a) (i) $m\ddot{x} = -\frac{\lambda x}{a}$

M1

$$= -amn^2 \frac{x}{a}$$

$$\ddot{x} = -n^2 x \quad \text{SHM}$$

A1

2

$$(ii) \quad T = \frac{2\pi}{n}$$

B1

1

$$(b) \quad (i) \quad m\ddot{x} = -amn^2 \frac{x}{a} - mkv$$

3 appropriate terms attempted

Signs consistent

M1

A1

$$\ddot{x} + k\dot{x} + n^2 x = 0$$

AG

A1

3

$$(ii) \quad p^2 + \frac{5n}{2}p + n^2 = 0$$

$$\left(p + \frac{5n}{4}\right)^2 - \frac{9n^2}{16} = 0$$

M1

$$p + \frac{5n}{4} = \pm \frac{3n}{4}, \quad p = -2n, \quad p = -\frac{n}{2}$$

$$2p^2 + 5np + 2n^2 = 0$$

$$(2p + n)(p + 2n) = 0$$

$$p = -\frac{n}{2}, p = -2n$$

A1

$$x = Ae^{-2nt} + Be^{-\frac{n}{2}t}$$

M1

$$t = 0, x = 0: A + B = 0$$

$$t = 0, \dot{x} = U$$

$$\dot{x} = -2nAe^{-2nt} - \frac{n}{2}Be^{-\frac{n}{2}t}$$

m1

$$U = -2nA - \frac{n}{2}B$$

$$A = -\frac{2U}{3n} \quad B = \frac{2U}{3n}$$

$$x = \frac{2U}{3n} \left(e^{\frac{-nt}{2}} - e^{-2nt} \right)$$

A1,A1

6



Accept sketch with correct shape not reaching origin but not crossing x -axis elsewhere

Accept reference to real distinct roots of auxiliary equation

B1

Heavy damping

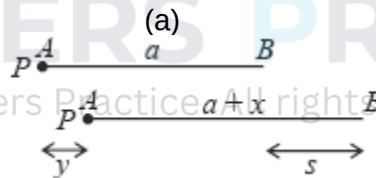
Independent of previous mark

B1

2

[14]

Q3.



$$s = \frac{1}{2}ft^2$$

B1

$$s = x + y \therefore y = \frac{1}{2}ft^2 - x$$

AG

M1A1

3

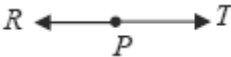
(b)

$$\frac{dy}{dt} = ft - \frac{dx}{dt}$$

AG

B1

1

(c)  $\frac{d^2 y}{dt^2} = f - \frac{d^2 x}{dt^2}$

B1F

$$F = ma : \frac{md^2 y}{dt^2} = \frac{2mn^2 ax}{a} - 3mn \left(ft - \frac{dx}{dt} \right)$$

M1M1A1

$$f - \frac{d^2 x}{dt^2} = 2mn^2 x - 3nft + 3n \frac{dx}{dt}$$

$$\frac{d^2 x}{dt^2} + 3n \frac{dx}{dt} + 2n^2 x = f + 3nft$$

AG

A1

5

(d) A.E. $p^2 + 3p + 2 = 0$
 $(p + 2)(p + 1) = 0$

M1

$$p = -1 \text{ or } p = -2$$

A1

EXAM PAPERS PRACTICE

$$x = Ae^{-2t} + Be^{-t} + \frac{f}{4} (6t - 7)$$

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M1A1

$$t = 0, x = 0 \Rightarrow 0 = A + B - \frac{7f}{4}$$

$$t = 0, \dot{x} = 0 \Rightarrow \dot{x} = -2Ae^{-2t} - Be^{-t} + \frac{6f}{4}$$

M1

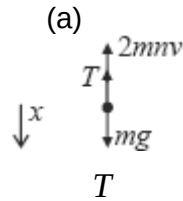
$$0 = -2A - B + \frac{6f}{4}$$

$$B = 2f, A = \frac{-f}{4}$$

A1,A1

$$x = 2fe^{-t} - \frac{f}{4}e^{-2t} + \frac{f}{4}(6t - 7)$$

Q4.



B1

$$m\ddot{x} = mg - \frac{5mn^2a}{a} \left(x + \frac{g}{5n^2} \right) - 2mn\dot{x}$$

all terms

M1A1A1

$$\ddot{x} + 2n\dot{x} + 5n^2x = 0$$

CAO; AG

A1

5

(b) AE: $p^2 + 2np + 5n^2 = 0$

M1

$$(p + n)^2 + 4n^2 = 0$$

$$p = -n \pm 2ni$$

A1

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$$x = e^{-nt} (A \cos 2nt + B \sin 2nt)$$

A1

$$t = 0, x = 0 \Rightarrow A = 0$$

A1

$$\dot{x} = -e^{-nt} (B \sin 2nt) + e^{-nt} (2nB \cos 2nt)$$

M1

$$t = 0, x = U \Rightarrow$$

$$U = 2nB \quad B = \frac{U}{2n}$$

A1

$$x = \frac{U}{2n} e^{-nt} \sin 2nt$$

A1

7

(c) light

B1

1

(d) $x = 0 \therefore \sin 2nt = 0$

M1

$$2nt = (0, \dots) \pi, \dots$$

$$t = \frac{\pi}{2n}$$

A1

$$\begin{aligned} \dot{x} &= -e^{-nt} \left(\frac{U}{2n} \sin 2nt \right) + e^{-nt} (U \cos 2nt) \\ &= -e^{-\frac{\pi}{2}} (0) + e^{-\frac{\pi}{2}} (U)(-1) \end{aligned}$$

M1

$$\text{speed} = Ue^{-\frac{\pi}{2}}$$

A1

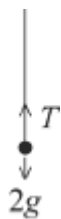
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EXAM PAPERS PRACTICE [17]

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Q5.

(a)



$$2g = \frac{49 \times e}{0.5}$$

M1

$$e = 0.2 \text{ metres}$$

A1

2

(b)



$$2\ddot{x} = 2g + F - T$$

M1A1

$$2\ddot{x} = 2g + 12 \cos nt - \frac{49(x + 0.2)}{0.5} \quad \text{Tension ft (a)}$$

B1F

$ft(a)$

A1F

$$\ddot{x} + 49x = 6 \cos nt$$

AG

A1

5

(c) $n = 5$ PI, $x = A \cos 5t + B \sin 5t$

Accept cos term only

M1

$$\dot{x} = -5A \sin 5t + 5B \cos 5t$$

A1

$$\ddot{x} = -25A \cos 5t - 25B \sin 5t$$

A1

Subs:

$$-25A \cos 5t - 25B \sin 5t + 49A \cos 5t + 49B \sin 5t = 6 \cos 5t$$

$$B = 0, \quad A = \frac{1}{4}$$

A1

A. eqn: $m^2 + 49 = 0 \quad m = \pm 7i$

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C.F.: $x = C \cos 7t + D \sin 7t$

A1

Gen sol: $x = C \cos 7t + D \sin 7t + \frac{1}{4} \cos 5t$

M1

$$t = 0, x = 0: \quad 0 = C + \frac{1}{4} \quad C = -\frac{1}{4}$$

A1

$$\dot{x} = -7C \sin 7t + 7D \cos 7t - \frac{5}{4} \sin 5t$$

m1

$$t = 0, \dot{x} = 0: D = 0$$

$$x = \frac{1}{4}(\cos 5t - \cos 7t)$$

A1

10

(d) Resonance occurs when $n = 7$

B1

1

[18]

Q6.

- (a) Natural length of AP is $4a$ and natural length of BP is $2a$
When particle is x from equilibrium position:

$$\text{Tension in } AP \text{ is } \frac{4mn^2a(2a+x)}{4a}$$

M1A1

$$\text{Tension in } BP \text{ is } \frac{4mn^2a(a-x)}{2a}$$

M1A1

In general position, using $F = ma$:

$$m \frac{d^2x}{dt^2} = \frac{4mn^2(a-x)}{2a} - \frac{4mn^2a(2a+x)}{4a} - 2mn \frac{dx}{dt}$$

M1A1

$$m\ddot{x} = 2mn^2a - 2mn^2x - 2mn^2a - mn^2x - 2mn\dot{x}$$

$$\frac{d^2x}{dt^2} + 2n \frac{dx}{dt} + 3n^2x = 0$$

AG

A1

7

- (b) [Substituting $x = Ae^{pt}$]
 $p^2 + 2p + 3 = 0$

M1A1

$$p = -1 \pm \sqrt{2}i$$

A1

General solution is:

$$x = e^{-t}(A \cos \sqrt{2}t + B \sin \sqrt{2}t)$$

A1

When $t = 0$, $x = \frac{1}{2}a \Rightarrow \frac{1}{2}a = A$

B1

Differentiating:

$$\frac{dx}{dt} = -e^{-t}(A \cos \sqrt{2}t + B \sin \sqrt{2}t) + e^{-t}(-A\sqrt{2} \sin \sqrt{2}t + B\sqrt{2} \cos \sqrt{2}t)$$

M1A1

When $t = 0$, $\frac{dx}{dt} = 0$
 $\Rightarrow 0 = -A + \sqrt{2}B$

$$A = \frac{1}{2}a, B = \frac{1}{2\sqrt{2}}a$$

A1

$$x = ae^{-t}\left(\frac{1}{2} \cos \sqrt{2}t + \frac{1}{2\sqrt{2}} \sin \sqrt{2}t\right)$$

8

[15]

Q7.

(a) Using $F = ma$,

$$2m\ddot{x} = 2mg - T - 4mnv$$

3 terms at least correct

M1

$$= 2mg - 8mn^2a \cdot \frac{x}{a} - 4mnv$$

A1

$$m\ddot{x} + 4mn^2x + 2mnv = mg$$

$$\text{i.e. } \frac{d^2x}{dt^2} + 2n \frac{dx}{dt} + 4n^2x = g$$

A1

3

(b) Substituting $x = Ae^{kt}$

$$p^2 + 2np + 4n^2 = 0$$

M1

$$p = \frac{(-2n \pm \sqrt{4n^2 - 16n^2})}{2}$$

$$= -n \pm \sqrt{3}in$$

A1

C.F. is

$$x = e^{-nt} (A \cos \sqrt{3}nt + B \sin \sqrt{3}nt)$$

Dep M1

B1ft

$$\text{P.I. is } x = \frac{g}{4n^2}$$

B1

General solution is $x =$

$$e^{-nt} (A \cos \sqrt{3}nt + B \sin \sqrt{3}nt) + \frac{g}{4n^2}$$

ft dep M1 first line (b)

M1

At least one of CF, PI correct

A1ft

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$$\text{When } t = 0, x = 0, \Rightarrow 0 = A + \frac{g}{4n^2}$$

M1

$$\therefore A = -\frac{g}{4n^2}$$

A1

$$\text{When } t = 0, \frac{dx}{dt} = 0;$$

$$\frac{dx}{dt} = -ne^{-nt} (A \cos \sqrt{3}t + B \sin \sqrt{3}t) +$$

M1

$$e^{-nt}(-\sqrt{3} nA \sin \sqrt{3} nt + \sqrt{3} nB \cos \sqrt{3} nt)$$

A1

$$\Rightarrow 0 = -nA + \sqrt{3} nB$$

$$\therefore B = \frac{1}{\sqrt{3}} A = \frac{g}{4\sqrt{3}n^2}$$

A1

$$\therefore x = \frac{g}{4n^2} - \left(\frac{g}{4n^2} \cos \sqrt{3} nt + \frac{g}{4\sqrt{3}n^2} \sin \sqrt{3} nt \right) e^{-nt}$$

A1

2

$$(c) \quad \frac{dx}{dt} = ne^{-nt} \left(\frac{g}{4n^2} \cos \sqrt{3} nt + \frac{g}{4\sqrt{3}n^2} \sin \sqrt{3} nt \right)$$

M1

$$-e^{-nt} \left(-\frac{\sqrt{3}g}{4n^2} \sin \sqrt{3} nt + \frac{g}{4n} \cos \sqrt{3} nt \right)$$

A1

$$\frac{dx}{dt} = 0 \text{ when}$$

$$\cos \sqrt{3} nt + \frac{1}{\sqrt{3}} \sin \sqrt{3} nt + \sqrt{3} \sin \sqrt{3} nt$$

$$- \cos \sqrt{3} nt = 0$$

M1

$$\sin \sqrt{3} nt = 0$$

$$\text{First time at rest is when } t = \frac{\pi}{\sqrt{3}n}$$

A1

4

[19]

Q8.

$$(a) \quad T = 10(x - 0.5) \text{ or } -10(x - 0.5)$$

M1A1

2

(b) $10(x - 0.5) - 2v = 1 \frac{dv}{dt}$
Newton's 2nd law

M1A1F

$$\frac{dv}{dt} + 2v - 10x = -5$$

AG

A1

3

- (c) (i) The rate of change of the length of the spring = speed of *A* relative to *B*
OE

E1

1

(ii) $\therefore \dot{x} = 1 - v$
 $-\ddot{x} + 2(1 - \dot{x}) - 10x = -5$

M1

$$\ddot{x} + 2\dot{x} + 10x = 7$$

AG

A1

2

- (d) Auxiliary equation: $m^2 + 2m + 10 = 0$

M1

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$$m = -1 \pm 3i$$

A1

CF $x = e^{-t} (A \sin 3t + B \cos 3t)$

M1

PI $x = \alpha$

$$\dot{x} = \ddot{x} = 0$$

M1

A1F

$$\alpha = \frac{7}{10}$$

GS $x = e^{-t} (A \sin 3t + B \cos 3t) + \frac{7}{10}$

A1F

6

(e) $t \rightarrow \infty$

$e^{-t} \rightarrow 0$

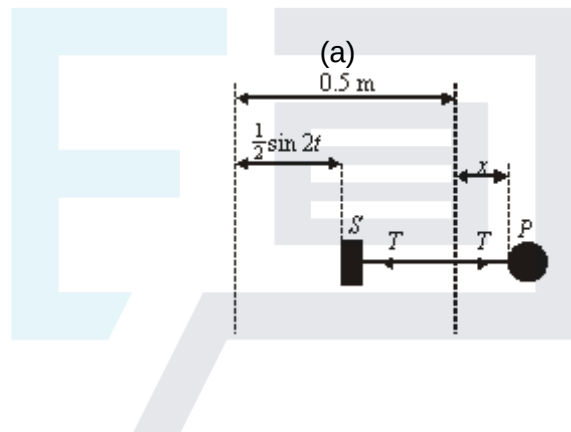
$x \rightarrow \frac{7}{10}$

M1A1F

2

[16]

Q9.



B1

1

(b) $T = 4.5 \left(\frac{1}{2} \sin 2t - x \right)$

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$T = 0.5 \ddot{x}$

A1

$0.5 \ddot{x} = 4.5 \left(\frac{1}{2} \sin 2t - x \right)$
Newton's 2nd law

M1

$\ddot{x} + 9x = \frac{9}{2} \sin 2t$
 AG

A1

4

(c) Auxiliary eqⁿ. $m^2 + 9 = 0$

M1

$$m = \pm 3i$$

A1

C.F. $x = A\cos 3t + B\sin 3t$

B1

For P.I. $x = p \cos 2t + q \sin 2t$

$$\begin{aligned}\dot{x} &= -2p \sin t + 2q \cos 2t \\ \ddot{x} &= -4p \cos 2t - 4q \sin 2t\end{aligned}$$

M1

$$5p \cos 2t + 5q \sin 2t = \frac{9}{2} \sin 2t$$

M1

$$\begin{aligned}p &= 0, q = \frac{9}{10} \\ \text{For both } p \text{ and } q\end{aligned}$$

A1F

G.S. is

$$x = A \cos 3t + B \sin 3t + \frac{9}{10} \sin 2t$$

A1F

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$$\begin{aligned}x &= \dot{x} = 0, t = 0 \\ \therefore A &= 0\end{aligned}$$

A1F

$$\begin{aligned}\dot{x} &= 3B \cos 3t + \frac{9}{5} \cos 2t \\ \therefore B &= -\frac{3}{5}\end{aligned}$$

A1F

$$x = -\frac{3}{5} \sin 3t + \frac{9}{10} \sin 2t$$

9

(d) $\dot{x} = -\frac{9}{5} \cos 3t + \frac{9}{5} \cos 2t$

M1

$$= -\frac{9}{5} \cos \frac{6\pi}{5} + \frac{9}{5} \cos \frac{4\pi}{5}$$

$$= 0$$

A1

2

[16]

Q10.

(a) $2 \times 9.8 = \frac{\lambda}{0.5} \times 0.2$

Equilibrium considered to form equation in λ

M1

$$\lambda = \frac{9.8}{0.2} = 49 \text{ N}$$

Correct equation

A1

Correct λ

A1

3

(b) $T = \frac{49x}{0.5} + 2g$

Equation for tension with two terms

M1

Correct equation

A1

$$2 \frac{d^2 x}{dt^2} = 2g - (98x + 2g)$$

$$\text{Use of } F = m \frac{d^2 x}{dt^2}$$

M1

$$\frac{d^2 x}{dt^2} = -\frac{98}{2} x = -49x$$

Correct equation

A1

ag Correct result from correct working

A1

5

(c) $\text{Period} = \frac{2\pi}{\sqrt{49}}$
Finding period

M1

Correct period

A1

$$t = \frac{1}{4} \times \frac{2\pi}{7} = \frac{\pi}{14} = 0.224 \text{ seconds}$$

Dividing period by 4

M1

Correct time

A1

4

[12]

Q11.

(a) $mg = \frac{\lambda}{l} \times \frac{l}{4}$

Consideration of forces in equilibrium

M1

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$$\lambda = 4mg$$

ag Correct λ from correct working

A1

2

(b) (i) $m \frac{d^2 x}{dt^2} = mg - T$
 $= mg - \frac{4mg}{l} \left(x + \frac{l}{4} \right)$

Two term expression for T

M1

Correct expression for T

A1

Three term equation of motion

M1

$$= -\frac{4mg}{l}x$$

Simplifying

m1

$$\frac{d^2x}{dt^2} = -\frac{4g}{l}x$$

ag Correct result from correct working

A1

5

$$(ii) \text{ Period} = 2\pi\sqrt{\frac{1}{4g}} = \pi\sqrt{\frac{1}{g}}$$

Identifying ω

Correct period

M1

A1

2

[9]

Q12.

(a) When displacement is x ,

$$\text{Tension} = \frac{\lambda x}{l} = \frac{4mn^2ax}{2a}$$

$$= 2mn^2x$$

B1

Using $F = ma$,

$$m = \frac{d^2x}{dt^2} = mg - 2mn^2x - 2mn\frac{dx}{dt}$$

M1 A1

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 2n^2x = g$$

A1

4

$$(b) \text{ CF } x = Ae^{pt} \\ p^2 + 2np + 2n^2 = 0$$

M1

$$p = \frac{-2n \pm \sqrt{4n^2 - 8n^2}}{2}$$

$$= -n \pm ni$$

A1

$$\therefore x = e^{-nt} (A \cos nt + B \sin nt)$$

A1

$$\text{PI } x = \frac{g}{2n^2}$$

B1

$$x = e^{-nt} (A \cos nt + B \sin nt) + \frac{g}{2n^2}$$

M1

When $t = 0$, $x = 0$,

$$A = -\frac{g}{2n^2}$$

B1

$$\frac{dx}{dt} = -ne^{-nt} (A \cos nt + B \sin nt) +$$

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$$e^{-nt} (-nA \sin nt + nB \cos nt)$$

$$\text{When } t = 0, \frac{dx}{dt} = 0$$

$$\Rightarrow -nA + nB = 0$$

$$B = -\frac{g}{2n^2}$$

A1

$$x = \frac{g}{2n^2} \{1 - e^{-nt} (\cos nt + \sin nt)\}$$

A1

Q13.

$$(a) \quad T_{AP} = \frac{\lambda \cdot 3a}{4a} = \frac{3}{4} \lambda$$

$$T_{PB} = 4mg \cdot \frac{a}{2a} = 2mg$$

Either

B1

Using $F = ma$ vertically

M1

$$mg + T_{PB} = T_{AP}$$

$$\therefore mg + 2mg = \frac{3}{4} \lambda$$

A1

$$\lambda = 4mg$$

A1

4

- (b) (i) When particle is moved a distance x below the equilibrium position, forces acting on it are

$$mg, T_{AP} = \frac{\lambda(3a+x)}{4a} = \frac{mg(3a+x)}{a},$$

M1

$$T_{PB} = 4mg \cdot \frac{(a-x)}{2a} = \frac{2mg}{a}(a-x)$$

$$\text{and resistance } \frac{1}{5}mk\dot{x}$$

All four forces

M1

[forces 2 and 4 are upwards]

Using $F = ma$ vertically downwards

$$m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{5}mk\dot{x}$$

Dependent on both M1 above

m1

$$m\ddot{x} = mg + \frac{2mg}{a}(a-x) - \frac{mg(3a+x)}{a} - \frac{1}{3}mk\dot{x}$$

A1

$$\ddot{x} - g - \frac{2g}{a}(a-x) + \frac{g(3a+x)}{a} + \frac{1}{3}k\dot{x} = 0$$

$$\ddot{x} + \frac{1}{3}k\dot{x} + \frac{3gx}{a} = 0$$

A1

$$10 \frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + 5k^2x = 0$$

A1

6

(ii) Substituting $x = Ae^{nt}$,

$$10n^2 + 2kn + 5k^2 = 0$$

$$n = \frac{-2k \pm \sqrt{4k^2 - 200k^2}}{20}$$

M1

$$= \frac{1}{10}(-k \pm 7ki)$$

A1

$$x = e^{-\frac{k}{10}t} (A \cos \frac{7}{10}kt + B \sin \frac{7}{10}kt)$$

M1 A1ft

$$\text{When } t = 0, x = \frac{a}{2}, A = \frac{a}{2}$$

B1

Differentiating

$$\frac{dx}{dt} = -\frac{k}{10}e^{-\frac{k}{10}t} (A \cos \frac{7}{10}kt + B \sin \frac{7}{10}kt) +$$

$$e^{-\frac{k}{10}t} (-\frac{7}{10}kA \sin \frac{7}{10}kt + \frac{7}{10}kB \cos \frac{7}{10}kt)$$

M1 A1ft

When $t = 0$, $\frac{dx}{dt} = 0$, $0 = \frac{k}{10} A + \frac{7}{10} kB$

M1

$$B = \frac{a}{14}$$

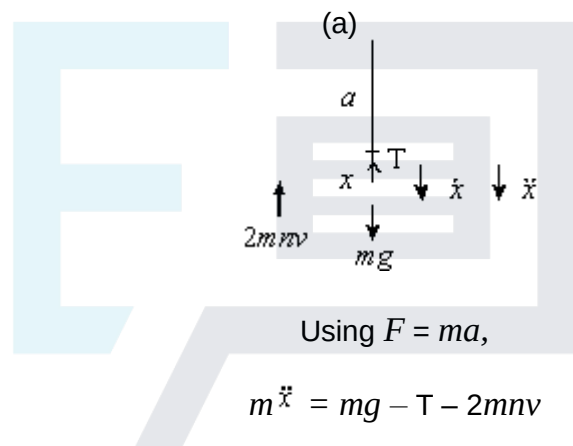
$$x = \frac{a}{14} e^{-\frac{k}{10}t} (7 \cos \frac{7}{10}kt + \sin \frac{7}{10}kt)$$

A1

9

[19]

Q14.



M1

$$= mg - 6mn^2 a, \quad \frac{x}{a} - 2mnv$$

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A1

$$m\ddot{x} + 6mn^2 x + 2mnx = mg$$

$$\text{i.e. } \frac{d^2x}{dt^2} + 2n \frac{dx}{dt} + 6n^2 x = g$$

A1

3

(b) Substituting $x = Ae^{pt}$

M1

$$p^2 + 2np + 6n^2 = 0$$

$$p = \frac{(-2n \pm \sqrt{4n^2 - 24n^2})}{2}$$

$$= -n \pm \sqrt{5}in$$

A1

$$\therefore \text{CF is } x = e^{-nt} (A \cos \sqrt{5} nt + B \sin \sqrt{5} nt)$$

A1

$$\text{PI is } x = \frac{g}{6n^2}$$

B1

$$\therefore \text{GS is } x = e^{-nt} (A \cos \sqrt{5} nt + B \sin \sqrt{5} nt)$$

$$= \frac{g}{6n^2}$$

A1

$$\text{When } t = 0, x = 0 \Rightarrow 0 = A + \frac{g}{6n^2}$$

M1

$$\therefore A = -\frac{g}{6n^2}$$

A1

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$$\text{When } t = 0, \frac{dx}{dt} = 0;$$

$$\frac{dx}{dt} = -ne^{-nt} (A \cos \sqrt{5} nt \times B \sin \sqrt{5} nt)$$

$$\left| + e^{-nt} (-\sqrt{5}n A \sin \sqrt{5} nt + \sqrt{5}n B \cos \sqrt{5} nt) \right|$$

$$+ e^{-nt} (-\sqrt{5}n A \sin \sqrt{5} nt + \sqrt{5}n B \cos \sqrt{5} nt)$$

M1

$$\Rightarrow 0 = -n A \times \sqrt{5}nB)$$

$$\therefore B = \frac{1}{\sqrt{5}} A = -\frac{g}{6\sqrt{5}n^2}$$

A1

$$\therefore x = \frac{g}{6n^2} -$$

$$\left(\frac{g}{6n^2} \cos \sqrt{5}nt + \frac{g}{6\sqrt{5}n^2} \sin \sqrt{5}nt \right) e^{-nt}$$

A1

10

$$(c) \quad \frac{dx}{dt} =$$

$$ne^{-nt} \left(\frac{g}{6n^2} \cos \sqrt{5}nt + \frac{g}{6\sqrt{5}n^2} \sin \sqrt{5}nt \right) -$$

$$e^{-nt} \left(-\frac{\sqrt{5}g}{6n} \sin \sqrt{5}nt + \frac{g}{6n} \cos \sqrt{5}nt \right)$$

M1

$$\frac{dx}{dt} = 0 \text{ when } \cos \sqrt{5}nt + \frac{1}{\sqrt{5}} \sin \sqrt{5}nt +$$

$$\sqrt{5} \sin \sqrt{5}nt \cos \sqrt{5}nt = 0$$

$$\sin \sqrt{5}nt = 0$$

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M1A1

$$\text{First time at rest is when } t = \frac{\pi}{\sqrt{5}n}$$

$$\text{When } t = \frac{\pi}{\sqrt{5}n}, \quad \frac{dx}{dt} = 0 \quad \text{SC2}$$

A1

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Examiner reports

Q2.

Candidates were able to attempt all parts of this question, but part (a) proved more challenging than expected, with some seeming unsure as to the requirements for simple harmonic motion. Part (b)(i) was done very well, as was part (b)(ii), although there were a number of errors in accuracy here. Part (b)(iii) proved very challenging; many successfully quoted the nature of the roots of the auxiliary equation as justification for heavy damping, but the hint of using a sketch was not appreciated.

Q3.

This question proved challenging for all candidates. The main difficulty was in appreciating the focus of the question being on the motion of the particle P , despite the hint in the early parts of the question. Part (a) was poorly answered, showing scant knowledge of the implications of forced motion. Despite this, part (b) was answered quite well, with some candidates realising that they needed to differentiate the answer from part (a). Part (c) was rarely answered well, due to reasons already mentioned, although many faked this solution. Most candidates were able to attempt the solution to the differential equation in part (d), but there were a variety of errors; some giving non-real solutions to the auxiliary equation, others ignoring the particular integral, or evaluating constants before including this.

Q4.

Part (a) was answered well, with occasional errors in finding the extension of the string. Part (b) was less successful, with some only being able to attempt the solving of the auxiliary equation before stopping. There were various algebraic errors, the most frequent being the omission of 'n' at varying stages, but still a number of good solutions. The answer to part (c) was well known. Part (d) proved quite testing in choosing correct methods for solution, and again algebraic errors marred solutions.

Q5.

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Part (a) was completed successfully by most candidates. Part (b) was often successfully answered, with a common error being the omission of the weight and an incorrect expression for the tension in the general position; often these two errors cancelled each other out in subsequent working, leading to an apparently correct solution. The main error in part (c) was to stop work having found the full general solution and not to evaluate the remaining constants. Part (d) was not answered well.

Q6.

Weaker candidates created the printed differential equation in part (a) by dubious means. It was necessary to find the equilibrium position and then use the extensions to be $2a + x$ in string AP and $a - x$ in string BP . Part (b) was answered very well by most candidates.

The only common error was in trying to insert the boundary conditions and finding $\frac{dx}{dt}$,
when $\frac{d}{dx} \cos \sqrt{2}t$ sometimes became $-\sin \sqrt{2}t$ rather than $-\sqrt{2} \sin \sqrt{2}t$.

Q7.

Part (a) was answered well, although again some candidates were clearly working back from the printed result to obtain the initial equation of motion. Part (b) was attempted well while the common error was in using $x = a$ when $t = 0$, not noticing that x was the distance below A , and initially the particle was released at A ; thus $\alpha = 0$ when $t = 0$. Relatively few candidates were totally accurate in finding the velocity of the particle in part (c) and only these successful candidates could notice that the terms in $\cos \sqrt{3nt}$ cancelled out, leaving only the terms containing $\sin \sqrt{3nt}$.

Q8.

The great majority of candidates were able to use Hooke's law and Newton's second law to answer parts (a) and (b) of this question. Some candidates were unable to explain why $\dot{x} = 1 - v$ for part (c)(i) but were still able to gain full marks for part (c)(ii).

Some candidates were able to answer part (d) of the question without answering part (c). Even though only the general solution for the differential equation was requested, some candidates proceeded to find the particular solution also. Many candidates were able to state the limiting value of x requested in part (e).

Q9.

A relatively large number of candidates had difficulty drawing the force/displacement diagram requested for part (a) of this question. A great majority of the candidates were able to answer part (b) correctly even when some of these candidates had not drawn a correct diagram.

The responses to part (c) were generally good, indicating a high level of knowledge and skills on the part of candidates in solving second order differential equations where the roots of the auxiliary equation are complex.

Part (d) of this question proved too difficult for some candidates.

Q10.

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The candidates did well with part (a) and for quite a number of candidates these were the only marks that they obtained on this question.

In part (b), the majority of the candidates totally ignored the weight of the sphere and produced invalid arguments. There were some candidates who could produce the required result without difficulty.

In the last part, there were quite a few candidates who found the period, but did not realise that the time needed was a quarter of the period.

Q11.

Part (a) of this question was done very well by almost all of the candidates. Part (b)(i) was very much more demanding and there were very few correct solutions. Most of the candidates did not find a two term expression for the tension in the string and did not include gravity when applying Newton's second law. In contrast, part (b)(ii) was done quite well, although some candidates did have difficulty simplifying their results.

Q12.

In part (a), a significant proportion of candidates worked back from the given result. Others used x as the position beyond the equilibrium position rather than as stated, and then had to 'lose' terms. In part (b), candidates made very good progress with many obtaining the correct result. The common error was forgetting an " n " in the

differentiation of one of the terms when finding $\frac{dx}{dt}$ to insert a boundary condition.

Other candidates could not simplify correctly the term $\frac{-2n \pm \sqrt{-4n^2}}{2}$.

Q13.

Most candidates answered part (a) correctly but a number did not use $mg + T_{PB} = T_{AP}$ and fiddled the algebra to give the printed result, for which they gained no marks.

In part (b), candidates had to find tensions in the two springs AP and PB , with extensions $(3a + x)$ and $(a - x)$ respectively, and then use $F = ma$ downwards, giving

$m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{5}mkx$. Without these equations candidates were not given credit for producing the printed differential equation.

All candidates used the printed equation given in part (b)(i) to find x in part (ii). A number of candidates lost k in a variety of places, but virtually all obtained a result similar to that required.

Q14.

Many candidates attempted to find the extension of the string in the equilibrium position, although in this question it was not relevant. These candidates had no mg force present when they found their differential equation. As this equation was printed on the question paper, candidates who then 'created' the mg force rarely scored marks in part (a). In part (b), candidates usually obtained the general solution $x = e^{-nt} (A \cos \sqrt{5}nt + B \sin \sqrt{5}nt)$, and most attempted to find the particular solution. They then showed the correct method for obtaining the actual solution. Those who correctly found the answer to part (b) often

substituted $t = \frac{\pi}{\sqrt{5}n}$ to show that the particle was at rest.

Only the better candidates differentiated their answer to show that the particle was at rest.